2024

Baystate Baby Birth Weights

PYCATS

Welcome To Our Presentation

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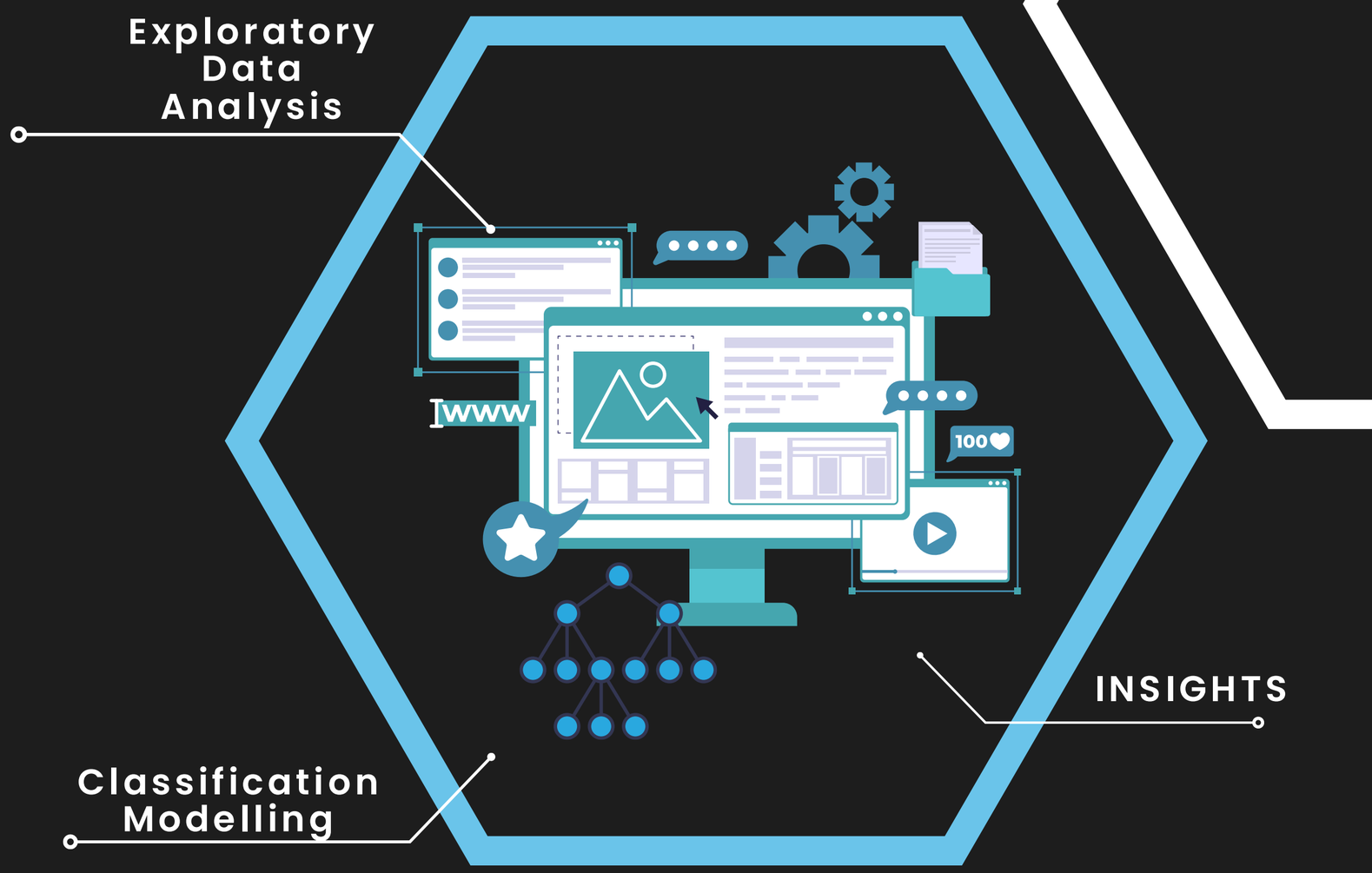
DATA ORIGIN AND OVERVIEW

BAYSTATE DATA SOURCE
https://www.key2stats.com/data-set/view/1131

- **Dataset Context:** Covers 189 newborn births at Baystate Medical Center, 1986.
- **Target Variable:** Focuses on 'low_birth_weight' as the primary outcome.
- **Risk Factors:** Includes maternal demographic and health-related factors.
- **Purpose:** Provides a basis for analyzing influences on neonatal weight.

Project Objective

- **Project Goal:** Analyze the relationship between infant birth weight and six predictors (three numerical, three categorical).
- **Analysis Objective:** Identify key factors contributing to "low" birth weight classification.
- **Methodology:** Utilize exploratory data analysis and classification modeling to evaluate variable impacts.
- **Outcome:** Determine significant contributors and identify the best machine learning model for prediction accuracy.



CHECK POINTS

189 OBSERVATIONS

TARGET VARIABLE

- low_birth_weight: indicator of birth weight less than 2.5kg

NUMERICAL VAR.

1. low_birth_weight
2. age_of_mother
3. last_menstrual_weight

CATEGORICAL VAR.

1. mother_race
2. smoking_status
3. hypertension_history
4. uterine_irritability
5. physician_visit
6. premature_labor

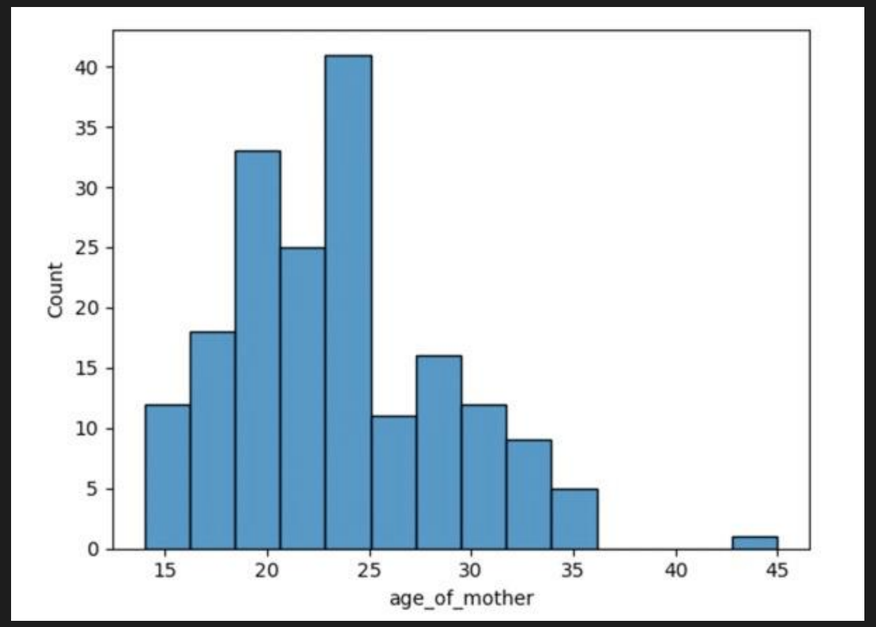
Data Prepocessing

- **Renamed variables for improved readability.**
- **Dropped irrelevant columns:**
  - birth_weight_grams: Used to create the boolean target variable.
  - weight: Unclear relevance to analysis.
- **Cleaned data for consistency:**
  - Checked numerical variables for non-numeric entries.
  - Ensured consistent labels in categorical variables (e.g., "Yes"/"No").
  - Removed 6 duplicate records.
- **Verified no missing values in the dataset.**
- **Retained outliers as they did not impact model performance in this small dataset.**
- **Mutated numerical column into new categorical variable**
  - age_category: Grouped mothers into 'Teen,' 'Young Adult,' and 'Mature Adult' age brackets.
- **Converted categorical variables to boolean datatypes using One-hot Encoding**

Exploratory Data Analysis: Numerical Variables

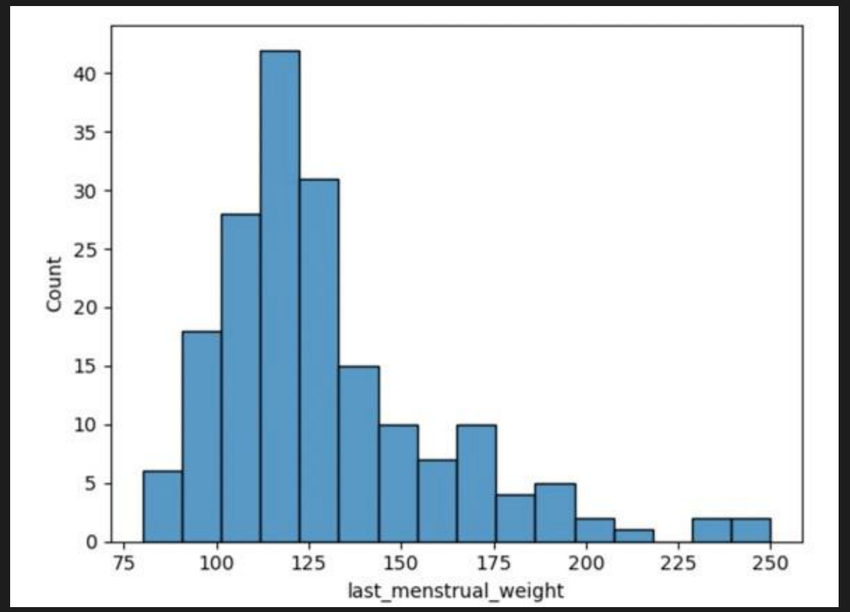
Age of Mother:

- Mean age: 23.42 years, Median: 23 years
- Age range: 14 to 45 years
- Most ages fall within 20 to 26.5 years, indicating a dataset primarily composed of younger mothers.
- Distribution is relatively symmetrical.



Last Menstrual Weight:

- Mean weight: 130.09 pounds, Median: 121 pounds
- Weight range: 80 to 250 pounds
- Distribution shows a slight right skew, suggesting higher weights are less common but present.



Exploratory Data Analysis: Categorical Variables

Low Birth Weight:

- 124 infants did not have LBW, 59 did.
- Indicates a significant absence of LBW among the infants studied.

Smoking Status:

- 70 mothers smoked during pregnancy, while 113 did not.
- Smoking is prevalent, impacting birth outcomes.

Mother's Race:

- Distribution: White (92), Other (66), Black (25)
- Majority are white, providing context for exploring racial disparities.

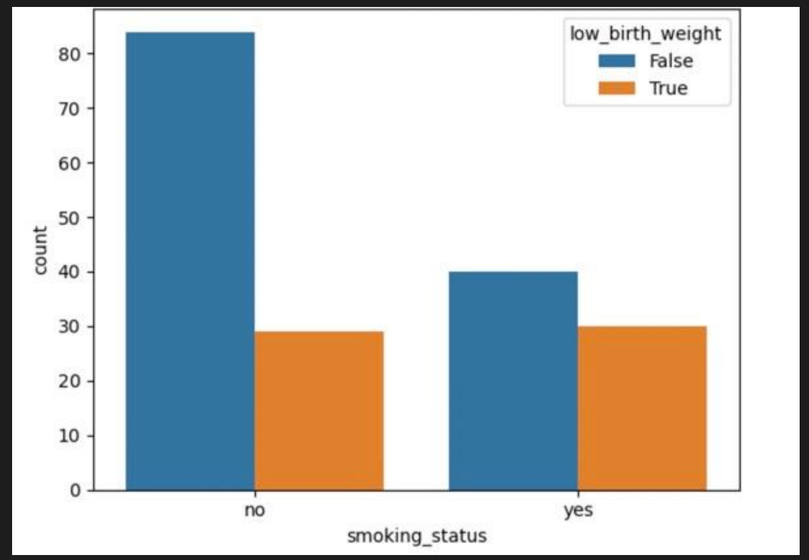
Health-Related Variables:

- Hypertension History: Only 12 out of 183 mothers reported hypertension.
- Uterine Irritability: Reported by 27 mothers.
- Premature Labors: Occurred in 30 mothers.
- Physician Visit: 88 mothers had visits in the first trimester; 95 did not.

Exploratory Data Analysis: Bivariate Insights

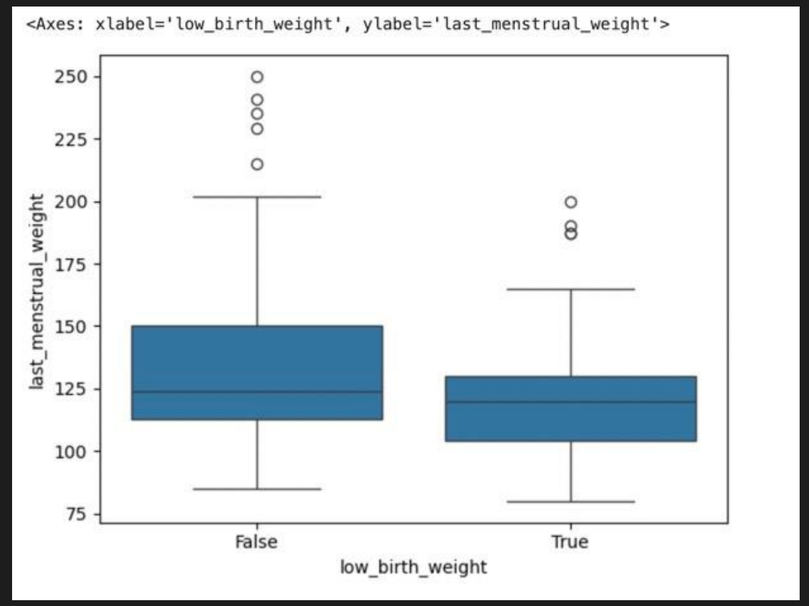
Impact of Smoking on Low Birth Weight:

- 42.86% of smoking mothers had LBW babies, compared to 25.66% of non-smoking mothers.
- Strong association between maternal smoking and LBW.



Last Menstrual Weight and Low Birth Weight:

- Mothers of LBW babies generally had lower last menstrual weights.
- Visualized through a box plot showing mean lines and distribution for LBW and non-LBW groups.



Machine Learning

Model selection Justification

Classification Analysis

Data Preprocessing part 2

One-HotEncoding ¶

```
[138]: # one-hot encoding
# pd.get_dummies(, drop_first=True)
df=pd.get_dummies(data=df, drop_first=True)
```

Training Process & Evaluation Metrics

```
[136]: # Make a prediction of y based on the optimal threshold, 0.34
pred_logr2 = pred_proba[:, [1]] >= 0.34
```

	precision	recall	f1-score	support
False	0.83	0.80	0.82	25
True	0.62	0.67	0.64	12
accuracy			0.76	37
macro avg	0.72	0.73	0.73	37
weighted avg	0.76	0.76	0.76	37

Model Selection

```
The best model is Logistic Regression with the Optimal Thresholds, 0.34 with f1_score of 0.64
```

```
The best model is Logistic Regression with the Optimal Thresholds, 0.34 with recall_score of 0.6666666666666666
```

Discussion: Model Selection

Rationale for Choosing Logistic Regression with Optimal Threshold 0.34

- High Performance: Achieved an F1-score of 0.64 and recall of 0.67, indicating strong predictive capability especially in minimizing false negatives.

- Optimal Balance: Threshold of 0.34 provided the best balance between precision and recall compared to other thresholds tested.

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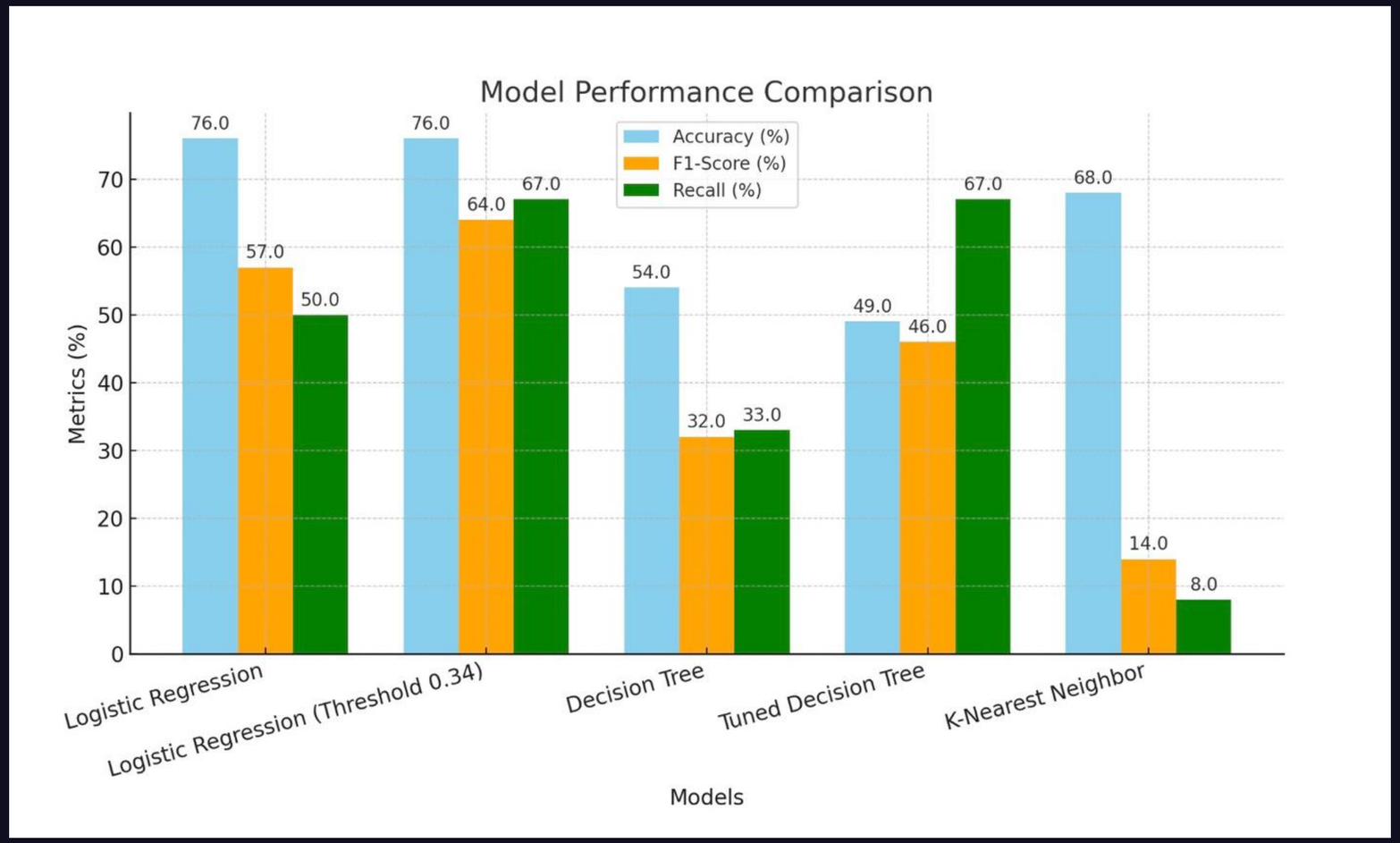
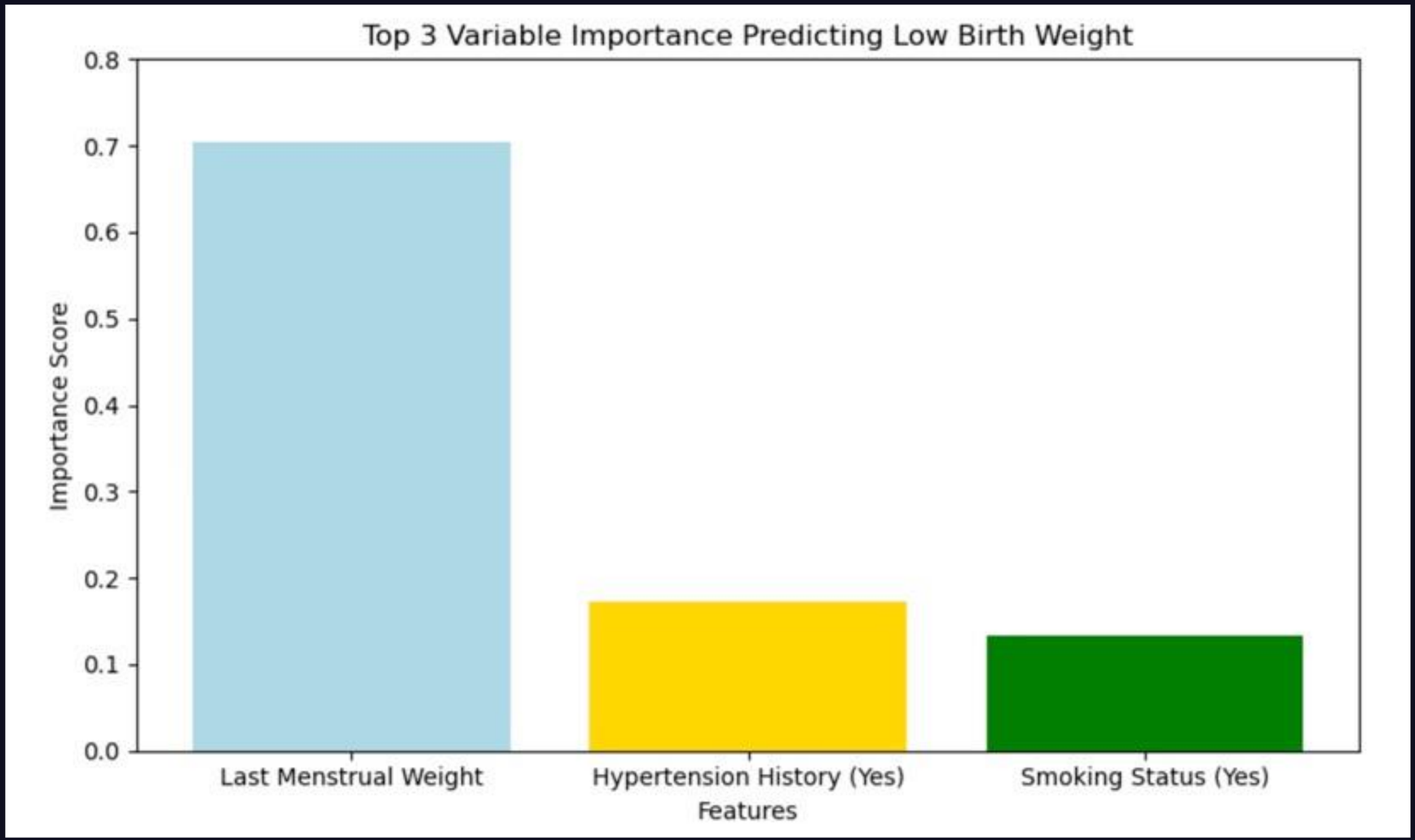
Comparison to other Supervised Machine Learning Model



Importance Insights



Last Menstrual Weight
Importance Score: *0.705*

Hypertension History (Yes)
Importance Score: *0.172*

Smoking Status (Yes)
Importance Score: *0.133*

Recommendations

- **Nutritional Support:** Ensure that expecting mothers receive adequate nutritional support to maintain a healthy weight throughout pregnancy. This can involve counseling on balanced diets and providing access to nutritional supplements if needed.

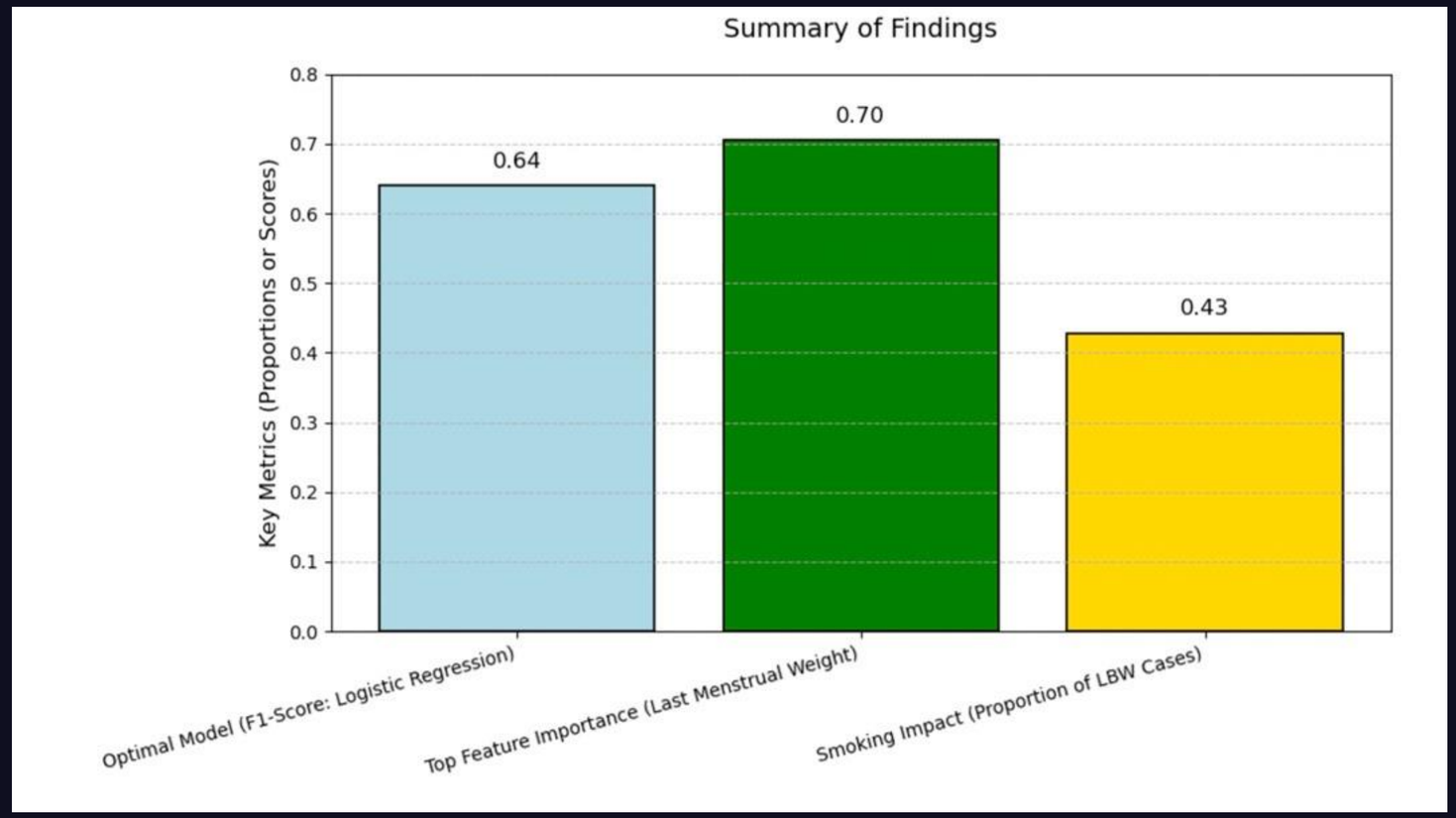
- **Regular Monitoring:** Implement regular weight monitoring during prenatal visits to detect and address any significant weight changes early.

- **Medical Management**: Provide appropriate medical management and treatment for pregnant women with a history of hypertension. This can include medications, lifestyle changes, and regular monitoring of blood pressure.
- **Education and Counseling**: Offer education on managing hypertension through diet, exercise, and stress reduction techniques.

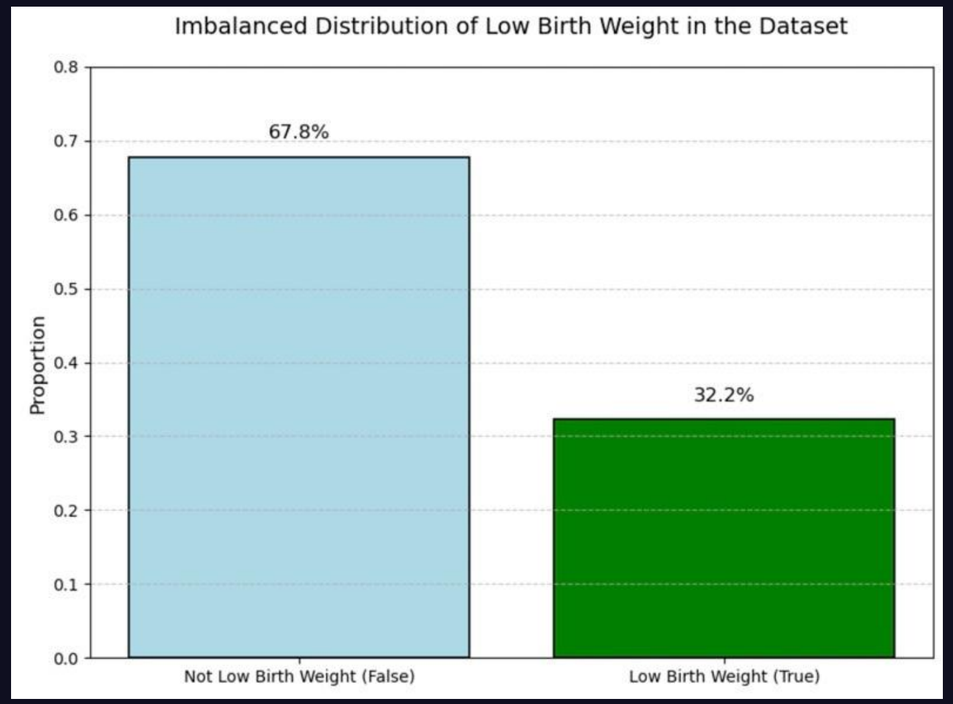
- **Smoking Cessation Programs**: Implement smoking cessation programs specifically tailored for pregnant women. These programs can include counseling, support groups, and nicotine replacement therapies.
- **Awareness Campaigns**: Launch awareness campaigns highlighting the risks of smoking during pregnancy and promoting the benefits of quitting.

Conclusions

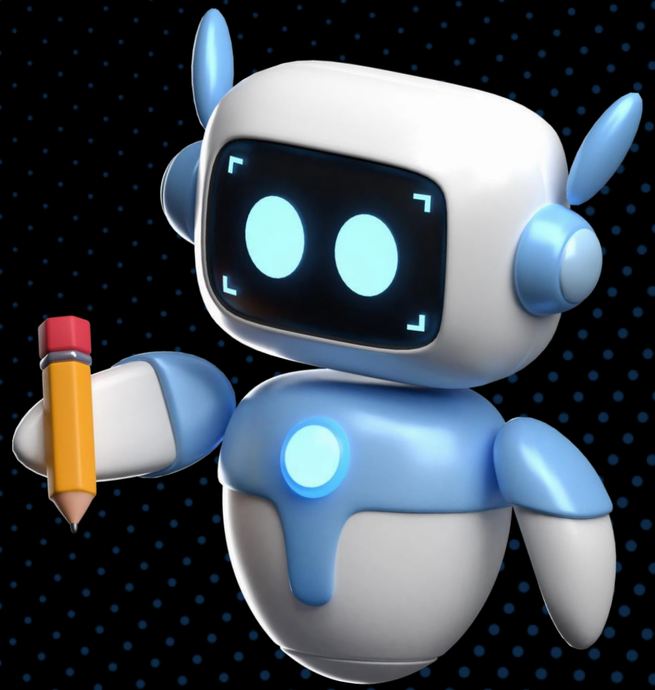
Summary of Findings



Project Limitations



THANK YOU!
&HAPPY HOLIDAYS



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